ACS Sustainable Chemistry & Engineering

Research Article

Cite This: ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

Techno-economic and Life Cycle Analysis for Bioleaching Rare-Earth Elements from Waste Materials

Vicki S. Thompson,[†] Mayank Gupta,[‡] Hongyue Jin,[‡] Ehsan Vahidi,[§] Matthew Yim,[†] Michael A. Jindra,[†] Van Nguyen,[†] Yoshiko Fujita,[†] John W. Sutherland,[§] Yongqin Jiao,[∥] and David W. Reed*[,†]

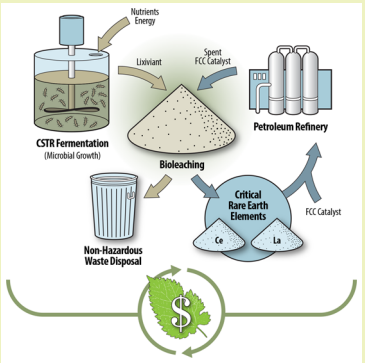
[†]Department of Biological and Chemical Processing, Idaho National Laboratory, P.O. Box 1625, Idaho Falls, Idaho 83415-2203, United States

[‡]School of Industrial Engineering, Potter Engineering Center, and [§]Environmental and Ecological Engineering, Potter Engineering Center, Purdue University, 500 Central Drive, Room 364, West Lafayette, Indiana 47907-2022, United States

[∥]Physical and Life Science Directorate, Biosciences and Biotechnology Division, Lawrence Livermore National Laboratory, 7000 East Avenue, Livermore, California 94550, United States

Supporting Information

ABSTRACT: A bioleaching process to extract rare-earth elements (REE) from fluidized catalytic cracking (FCC) catalysts was optimized using a heterotrophic bacterium *Gluconobacter oxydans* to produce organic acids from glucose. Parameters optimized included agitation intensity, oxygen levels, glucose concentrations, and nutrient additions. Biolixiviants from the optimized batch process demonstrated REE leaching efficiencies up to 56%. A continuous bioreactor system was subsequently developed to feed a leach process and demonstrated leaching efficiencies of 51%. A techno-economic analysis showed glucose to be the single largest expense for the bioleach process, constituting 44% of the total cost. The bioleaching plant described here was found profitable, although the margin was small. Lower cost carbon and energy sources for producing the biolixiviant, sourcing FCC catalysts with higher total REE content (>1.5% by mass), and improved leaching efficiencies would significantly increase the overall profit. A life cycle analysis showed that electricity and glucose required for the bioreactor had the largest potential for environmental impacts.



KEYWORDS: *Bioleaching, Rare-earth elements, Techno-economic assessment, Life cycle analysis, End-of-life products*

INTRODUCTION

Rare-earth elements (REE), which include the lanthanides together with yttrium and scandium, are essential for many of the technologies that underlie modern communications, renewable energy production, and advanced transportation systems.[1−3] China dominates global REE production and has been increasing its own use of REE, leaving the United States (U.S.) and other countries susceptible to supply disruptions.[4] Because of this vulnerability, the possibility of recovering and reusing REE from industrial wastes and end-of-life consumer products has recently become the subject of increased interest within the U.S. and internationally.[1,5−7]

Existing REE extraction methods generally rely on hydrometallurgical or pyrometallurgical approaches, sometimes combined with electrometallurgical methods.[1,8] However, in cases where these approaches are not economically or environmentally viable, such as when target metal concentrations in the feedstocks are low, or when generation of secondary hazardous waste streams are of concern, leaching using microorganisms or microbiological products may be an attractive alternative.[9−11] Heterotrophic organisms can produce organic acids from organic carbon sources and these acids have been applied to metal leaching.[12−22] In addition to their role in promoting dissolution of the solids due to acidity, some of the organic acids also have chelating properties that promote REE solubility.

Previously, we reported on our studies of REE leaching from fluorescent lamp wastes and spent petrochemical fluid catalytic cracking (FCC) catalysts using heterotrophic microbial cultures.[18] We found that heterotrophic bioleaching could indeed successfully extract metals from various waste feedstocks, using either spent culture or cell-free supernatants. Moreover, the microbially produced lixiviants (predominantly gluconic acid) were more effective at solubilizing REE than abiotically prepared solutions with higher gluconic acid concentrations, likely due to the biological production of additional organic acids that lowered the pH of the medium.

Subsequent to earlier studies, we have continued working with the microbial culture that produced the most effective lixiviants to optimize cultivation and leaching conditions with the goals of increasing acid production and improving REE leaching efficiency from FCC catalysts. The optimized

Received: August 11, 2017
Revised: December 5, 2017
Published: December 13, 2017

Downloaded via WORCESTER POLYTECHNIC INST on September 3, 2020 at 13:19:40 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

© 2017 American Chemical Society

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

protocols and parameters were used as the basis for a preliminary techno-economic analysis (TEA) and life cycle analysis (LCA) of the process, which together enable assessment of the potential for applying bioleaching to practical recovery of critical metals from industrial and postconsumer wastes.

MATERIALS AND METHODS

***Gluconobacter* Growth and Bioleaching Optimization.** *Batch Growth of Gluconobacter Strain and Preparation of Biolixiviant.* *Gluconobacter oxydans* strain NRRL B58[23] was obtained from the Agriculture Research Service, USDA (Peoria, IL). The Sixfors reactor system (Infors HT, Laurel, MD) was used for the batch growth studies. The reactors were inoculated with *G. oxydans* grown in modified Pikovskaya phosphate medium (Pkm) containing 10 g/L glucose.[18,24] A Pkm version with tryptone (PkmT, 10 g/L) was used in some cases. For optimization experiments, the reactors were operated with PkmT or Pkm containing glucose between 5 and 80 g/L, agitation rates from 200 to 1200 rpm, and air addition rates from 0.55 to 2.2 vessel volumes per minute (vvm) for roughly 36−40 h prior to collection of the culture. The cultures were harvested by centrifugation (6000*g*, 30 min) to settle cells and the supernatant was filtered (0.22 μm pore size) to produce the biolixiviant prior to storage at 4 °C.

Continuous Production of Biolixiviant and Testing of Different Dilution Rates. The Sixfors system was also used in a continuous stirred tank reactor (CSTR) mode for production of biolixiviant. After initial reactor inoculation, the *G. oxydans* cultures were grown under batch conditions for 15 h to reach mid log growth phase. Continuous culture was initiated by feeding Pkm amended with glucose (40 g/L) at various dilution rates (h^{-1}): 0.05, 0.16, 0.25 and 0.38. Dissolved oxygen and pH were continuously monitored and samples (~1.5 mL) were drawn from each culture periodically for measurement of optical density and organic acid production. The CSTRs were operated for over 100 h of continuous substrate feed and biolixiviant production.

REE Leaching. Spent FCC catalyst containing the key REE components cerium (Ce) and lanthanum (La) was obtained from Valero (Houston, TX). The REE composition and characterization of the FCC catalyst used in these experiments were reported previously;[18] the total REE content of the FCC catalyst was approximately 1.5% by mass.

Leaching studies were conducted using filtered biolixiviant with FCC catalyst. Prior to leaching, the solids were autoclaved three times to mitigate biological contamination. Leaching tests were performed over a range of pulp densities from 1.5 to 50% (solid-to-liquid mass ratios, w/w), in 50 mL conical tubes incubated 24 h at 30 °C with shaking at 150 rpm. Control treatments included FCC catalyst incubated with sterile Pkm medium without added glucose.

To test the effects of *G. oxydans* cells on leaching, leaching studies were also performed with biolixiviants that were not filtered to remove the bacterial cells prior to contact with the FCC. These studies were conducted in the Sixfors reactors operated under batch conditions as described above except that FCC catalyst was added directly to the culture after 12 or 36 h, and the reactors were operated for an additional 24−48 h.

Simulated heap leaching studies were also conducted to assess the effects of nonideal mixing that would be expected under those conditions. FCC catalyst was placed in a fritted glass Büchner funnel and filtered biolixiviant was pumped on top of the catalyst bed and allowed to percolate through the solids. The leachate was collected and pumped back onto the catalyst bed to form a recycle loop which operated for 24 h. The solid-to-liquid mass ratios tested were 1.5% and 50%, where the ratios represent the mass of the solid compared to the total volume of lixiviant applied, regardless of recycling.

Analytical Methods. Organic acid analysis was performed by high-performance liquid chromatography as described previously.[18] Gluconic acid (Sigma-Aldrich, St. Louis), 2-keto gluconic acid (Sigma-Aldrich), 5-ketogluconic acid (Carbosynth, Berkshire, U.K.), and 2,5-diketogluconic acid (provided by Masaaki Tazoe and Tatsuo Hoshino; NRL Pharma, Inc., Kawasaki, Japan) were used as standards.

REE concentrations were measured using ICP-MS. The ICP-MS instrument (Agilent 7900 with UHMI) was operated in accordance with manufacturer instructions. The filtered (0.22 μm Millex-GP PES) samples and the commercial standard stock solutions were acidified with ultrapure concentrated nitric acid to a concentration of 1% HNO_3 (v/v).

Techno-economic Analysis. *Raw Materials and Utilities.* The bioleaching plant was assumed to be capable of processing 18,838 ton of FCC catalyst feedstock annually or 10% of the FCC catalyst used in the United States[25,26] and located next to the existing FCC recovery infrastructure to minimize transportation and material handling costs.

The bioleaching process consisted of two unit operations: a bioreactor to produce organic acid (biolixiviant) which was then fed onto a leaching pile (the second unit operation) of FCC catalyst. The bioreactor for organic acid production was based on a proposed succinic acid production plant[27] and included unit operations for nutrient medium sterilization and cooling and aerobic batch fermentation (Figure S1, Supporting Information). The downstream separation and purification unit operations described for succinic acid were not included since purification of the organic acids produced was not necessary for lixiviant production. The main utilities required for the bioreactor operation included water, steam, and electricity. Water was used for the nutrient feed cooler, the heat exchanger, and bioreactor cooling as well as to make up the nutrient feed, while steam was required for sterilization of the nutrient feed and the bioreactor. It was assumed that the FCC catalyst would not be sterilized. Electricity was estimated based on the energy required for the operation of major equipment, such as bioreactor tank agitation, pumps for transfer of nutrient feed to the bioreactor and transfer of biolixiviant from the bioreactor to the leaching pile, and air compressors for bioreactor aeration.

Fixed Capital Costs. Fixed capital costs include equipment purchase and installation costs, piping costs, electrical equipment and material costs, and costs of instrumentation and control. Major equipment involved in the bioleaching process were the bioreactor, air compressor, nutrient feed sterilizer, cooler, and heat exchanger. The size of each piece of required equipment was estimated based on the target processing rate of spent FCCs as well as the required pulp density. Costs including installation and capital were scaled to capacity and calculated as described elsewhere.[27−29] Pulp densities of 1.5, 18, and 50% (w/w) were considered for batch-produced gluconic acid and a continuous production case with 50% pulp density was also examined.

Other Associated Costs. Other costs included labor, maintenance, administration, marketing, and research and development. These costs were divided into different categories: other direct costs (besides direct costs from raw material, and utilities), indirect costs, and general costs. Silla (2003) provides the empirical relations to estimate these costs (Table S1). Additional assumptions related to the plant operation, debt financing, income tax, and depreciation are listed in Table S2.

Revenue. There were assumed to be two sources of revenue from bioleaching of spent FCC catalysts: (1) fee charged to oil refineries ($200/ton) for disposal of FCC catalyst as a hazardous waste[30] and (2) sales of REE leachate. The bioleaching output is an aqueous solution that contains a mixture of REE products whose value is not well documented in the literature as open market prices are only available for >99% separated REE. An estimated price for mixed REE was developed using a techno-economic study performed by SRK Consulting on Mountain Pass Mine and its processing facilities.[26] From this study, the costs to leach REE from ore to produce a mixed REE stream were approximately 40% of the total REE production cost. The remaining 60% of the costs were for processes such as solvent extraction to concentrate and separate individual REE. On the basis of this, it was assumed that additional concentration and separation processes would be required to purify the bioleached mixed REE stream and, to cover this cost, the bioleached mixed REE stream value would be discounted to 40% of the market price.

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

Life Cycle Analysis. *System Boundary and Functional Unit.* The scope of this LCA study is a gate-to-gate analysis of the bioleaching process, starting with spent FCC catalyst and ending with a mixed aqueous REE stream. A schematic of the system boundaries is shown in Figure S2 which includes all of the energy and raw material consumption and translates all process data into specific influences on human health and the environment.

Emissions from the bioreactor itself (CO_2 from microbial respiration and any volatile compounds produced as part of metabolism) were assumed to be negligible compared to other emission sources. Although FCC catalysts accrue hazardous materials such as nickel, antimony, vanadium, and lead during their operational lifetime,[31] we did not consider the costs associated with hazardous waste disposal since that would be outside of the system boundaries for this analysis. However, as a conservative estimate, it was assumed that the net environmental liability (i.e., environmental credits for removing hazardous materials via bioleaching minus the environmental burden for disposing nonhazardous leach residue) was zero.

Life Cycle Inventory Analysis. Major energy and material inputs to process 1 ton of spent FCC catalyst at a bioleaching facility were estimated from Reed et al.[18] and Efe et al.[27] Water and steam consumption were calculated based on stoichiometry and heat-exchange rate. Inventory analysis was performed using Ecoinvent 3.0 database and SimaPro 8 software based on attributional modeling and uses the system model Allocation, Default (Alloc. Def.).

Impact Assessment. The Tool for the Reduction and Assessment of Chemical and other environmental Impacts (TRACI)[32] developed by the U.S. Environmental Protection Agency was employed for life cycle impact assessment as a midpoint-oriented technique for global warming (kg CO_2 equiv), eutrophication (kg N equiv), smog (kg O_3 equiv), ecotoxicity (CTUe), respiratory effects (kg $PM_{2.5}$ equiv), ozone depletion (kg CFC-11 equiv), carcinogenics (CTUh), noncarcinogenics (CTUh), acidification (kg SO_2 equiv), and fossil fuel depletion (MJ surplus).

RESULTS AND DISCUSSION

Optimization of Organic Acid Biolixiviant Production. *G. oxydans* cultivation was optimized for low-pH organic acid production by varying reactor operating conditions (agitation rate and oxygen addition) and growth medium constituents (tryptone and glucose). The testing showed that an agitation rate of 600 rpm and an air flow rate of 2.2 vvm resulted in the highest growth rate of *G. oxydans*, likely due to removal of oxygen limitations. Gluconic acid production and medium pH as a function of initial glucose concentration using those agitation and air flow conditions are shown in Figure 1.

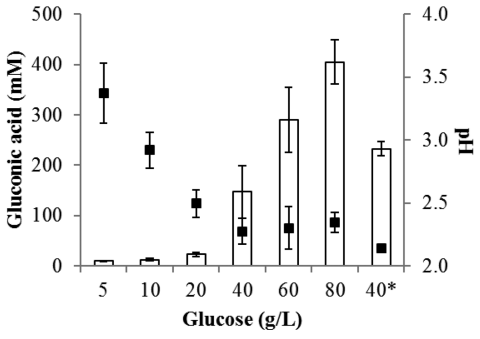


Figure 1. Biolixiviant from *G. oxydans* grown in reactors with 500 mL Pkm media containing tryptone. Gluconic acid concentrations (bars) produced by *G. oxydans* in medium amended with different initial amounts of glucose. Final pH (squares) of the medium following growth of *G. oxydans* with different concentrations of glucose. * Indicates Pkm media without tryptone. Error bars indicate standard deviations of 4−7 replicate cultures.

In the Pkm medium with tryptone the lowest average pH value of 2.3 was observed at 40 g/L glucose, although higher gluconic acid concentrations were produced when more glucose was provided. This is likely due to the production of other organic acids besides gluconic acid by *G. oxydans*. The further oxidized products keto-gluconic acid and 2,5 diketo-gluconic acid have pK_a values of 2.66 and 2.52, respectively, compared to 3.86 for gluconic acid. The removal of tryptone from the 40 g/L glucose medium increased average gluconic acid production to 233 mM, compared to 147 mM with tryptone, and resulted in an even lower pH of 2.14.

Given that batch gluconic acid production takes up to 40 h per batch and is associated with considerable downtime to prepare for the next batch, continuous production was examined as a strategy to provide organic acids in a quantity and at a rate useful for commercial bioleaching. The Pkm with 40 g/L glucose was used as the influent to the CSTRs. Decreased cell growth rates were observed in the batch reactors after about 16 h as cells entered stationary growth phase; however, in continuous culture sustained cell growth and organic acid production were maintained for over 100 h for the dilution rates tested (0.05−0.38 h^{-1}). Overall, the trend showed increasing acidity with decreasing dilution rates, with the lowest dilution rates 0.05 and 0.16 h^{-1} yielding the highest gluconic acid concentrations and lowest pH (Table 1). A small amount of 2,5-diketogluconic acid (8 ± 6 mM; data not shown) was detected only for the 0.05 h^{-1} dilution rate, likely because the pH of the bioreactors was not maintained within the optimal window for the oxidative conversion pathway from gluconic acid to 2,5 diketo-gluconic acid.[33]

Table 1. Effects of Dilution Rate (D) on *G. oxidans* Continuous Cultivation and Biolixiviant Production Using Pkm Medium with 40 g/L Glucose[a]

dilution rate (h^{-1})	gluconic acid (mM)	pH	cell density ($\times 10^7$/mL)	REE leached[b] (%)
batch[c]	233 (14)	2.14 (0.01)	79 (16)	49 (<1)
0.05	184 (16)	2.19 (0.03)	26 (7)	51 (4)
0.16	174 (12)	2.22 (0.06)	33 (5)	49 (5)
0.25	116 (23)	2.41 (0.06)	27 (7)	46 (3)
0.38[d]	68 (11)	2.61 (0.03)	26 (6)	42 (2)

[a]Values are averages (n > 3), with standard deviations shown in parentheses. [b]Percent of REE (predominantly La) solubilized per total REE available in 1.5% solid FCC catalyst. [c]Average values from the Pkm batch system at harvest (36−40 h). [d]Average values with n = 2.

Bioleaching Results. The effectiveness of the batch-produced biolixiviants was evaluated using FCC catalyst at pulp densities up to 50% (Figure 2). The highest extraction efficiencies were observed at the lowest pulp density tested (1.5%). This is not unexpected since mixing efficiencies were higher at lower solids loadings. Increased pulp densities resulted in lower extraction efficiencies, but higher total REE recovery for the same amount of biolixiviant used. Control leaching experiments with sterile Pkm indicated negligible leaching by the medium without prior *G. oxydans* cultivation.[18]

The leaching results discussed to this point were from microorganism-free spent medium. However, for an industrial process, harvesting cultures and removing microorganisms would be costly. Therefore, we examined leaching efficiency with microorganisms present in the Sixfors reactors and

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

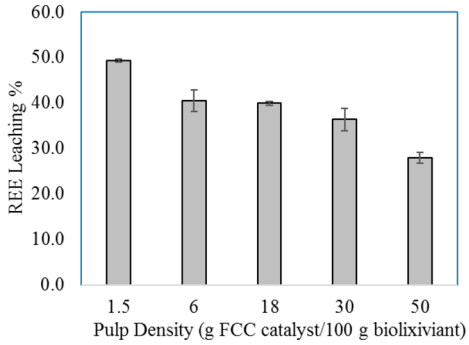


Figure 2. Percentage of total REE leached from the FCC catalyst after incubating for 24 h with different ratios of solid to biolixiviant. Biolixiviant was prepared from batch cultures of *G. oxydans* grown for 36–40 h. Error bars indicate standard deviations of triplicate experiments. Solubilization of REE by sterile Pkm media incubated with FCC catalyst was previously shown to be negligible.[18]

observed an increase in REE recovery efficiency, from 49% to >56% extraction of REE from 1.5% FCC catalyst. While this increase may have been due to improved agitation or contact of the biolixiviant with the solid, it is also possible that microorganism adsorption to the FCC catalyst surface reduced mass-transfer limitations by producing biolixiviant directly on the surface of the catalyst.

To assess whether leaching efficiency was altered for biolixiviant produced continuously versus that under batch conditions, biolixiviants produced in the CSTRs at different dilution rates were used to leach FCC catalyst. The observed REE leaching efficiencies for all of the CSTR-produced lixiviants were similar to the efficiency obtained for the batch lixiviant (Table 1), indicating that switching from batch to lower cost continuous production of lixiviant should not be detrimental to leaching performance.

Industrial bioleaching processes occur using a heap format as it becomes difficult and expensive to provide adequate mixing as pulp densities increase. To assess how this system would perform in a heap format, a bench-scale simulated heap system with recycling was tested. Biolixiviant was pumped onto the surface of a bed of FCC catalyst contained in the fritted glass Büchner funnel. The leachate was recycled and pumped back onto the catalyst bed for 24 h. At 1.5% pulp density, 46% of the REE was leached from the heap FCC catalyst compared to 49% from the batch reactor experiment. At 50% pulp density, 31% REE was leached from the heap FCC catalyst compared to 28% from the batch conical tube. These results indicate that a recycle heap leaching system can obtain comparable leach efficiencies to a mixed system.

Techno-economic Analysis. Three pulp densities were initially considered as base cases for the TEA: 1.5%, 18%, and 50% which had leaching efficiencies of 49%, 40%, and 28%, respectively. Although it might be expected that higher leaching efficiencies would be more economical, the bioreactor volumes required to produce the necessary biolixiviants at 1.5% and 18% (6436 and 536 m^3, respectively) were much higher than those for the 50% case (193 m^3). In addition to higher bioreactor costs, TEA results revealed higher costs associated with growth nutrients and utilities which significantly outweighed the higher leaching efficiencies; therefore, the 50% pulp density case was selected to serve as the base case (Table S3).

Table 2 shows the cost breakdown of the proposed bioleaching process. The major cost driver was identified to

Table 2. Total Cost Breakdown of Bioleaching

categories	cost ($/year)	cost (%)
nutrients	1,340,000	44.3%
electricity	304,000	10.0%
utility	181,000	5.99%
labor	215,000	7.11%
maintenance	114,000	3.77%
fixed capital (annualized)	80,600	2.66%
indirect	178,000	5.88%
general	613,000	20.2%

be the nutrient cost (constituting 44% of the total cost), within which glucose (the carbon source for *G. oxidans*) made up almost 98% of the total cost. The next two major costs are general costs at 20% and electricity at 10% of total cost. Tables S4 and S5 show more detailed information for costs in Table 2. The identification of the major contributors to the bioleaching costs helps to guide the direction of future efforts to reduce costs and earn higher profits. For example, the replacement of glucose by alternative substrates such as food-processing wastes or crop residues would substantially improve the economic outlook for this technology.

The total annual cost for the proposed bioleaching plant is $3,030,000. With the assumptions listed in Table S2, the payback period is estimated to be 1 year, after which the net cash flow becomes positive. The internal rate of return is 44% and net present value is $5,780,000 with a discount rate of 8%. With use of the 2016 REE prices, the total revenue stream was $3,880,000, of which $116,000 (Table S6 for individual REE contributions and 2016 prices) was from REE sales (the balance is the $200/ton tipping fee as discussed in the Techno-economic Analysis: Revenue section above). The annual profit from the bioleaching plant would be $855,000, although this number would be higher if the sale of other metals was considered (e.g., Al, Ni, Sb, V, and Si). This demonstrates that with current REE prices it will be important to capture the tipping fee for the spent FCC catalyst. However, the scenario could change dramatically if REE prices increase significantly. For example, if REE prices were at the values observed in 2011 (Table S6), the REE revenue would increase to $5,520,000/year which would make the plant profitable even without a tipping fee.

For comparison with traditional metallurgical processes, several chemical leaching processes for REE recovery from FCC using various mineral acids were also examined, based on conditions and recoveries reported in the literature (Table 3). Similar assumptions for costs were made for these processes as compared to bioleaching with the biology-specific costs such as nutrients, medium, and reactor sterilization and aeration removed. In addition, the same mass of FCC catalyst (18,838 ton/year) was assumed to be leached for these cases so that direct comparisons could be made. Since the output of these processes was also a mixed REE stream and further separation and purification would also be required,[26] the REE revenue was also discounted to 40% of market value. Many of these processes are not economical primarily due to the low pulp densities. One process described by Gao and Owens[34] with high pulp density and a shorter leach time has a relatively similar annual profit to the bioleaching case examined here.

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602–1609

Table 3. Profit Comparison for Leaching Alternatives for Extracting REEs from Spent FCC

sources	process settings	profits ($/year)
bioleaching	50% pulp density, 24 h leach time	855,000
Innocenzi et al.[35]	25 °C, 2 M H_2SO_4, 15% pulp density, 3 h leach time	−7,100,000
	80 °C, 2 M H_2SO_4, 15% pulp density, 3 h leach time	−7,750,000
Gao and Owens[34]	75 °C, 5% HCl, 25% pulp density, 2 h leach time	−407,000
	82 °C, 4% HNO_3, 31% pulp density, 30 min leach time	1,130,000
Zhao et al.[36]	2 M HCl, 10% pulp density, 45 °C, 4 h leach time	−8,820,000
	2 M HNO_3, 10% pulp density, 45 °C, 4 h leach time	−12,000,000
	1 M H_2SO_4, 10% pulp density, 45 °C, 4 h leach time	−5,460,000

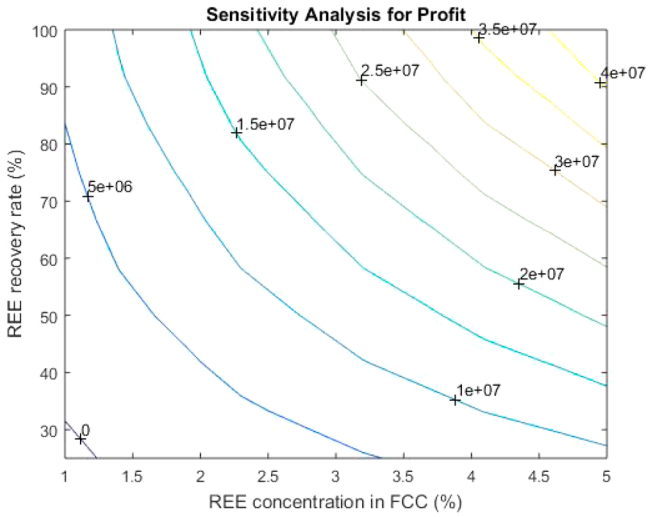


Figure 4. Sensitivity analysis on REE concentration and recovery rate. Contours represent annual profit in $/year.

Sensitivity analyses were performed for several key parameters that significantly affected the profitability of bioleaching. The first was glucose purchase cost (Table 2 and Table S3), which constituted 98% of the total nutrient costs and 44.3% of the total annual plant costs. A range of glucose purchase costs were examined ($0.40/kg to $0.87/kg), the latter of which is the bulk purchase price reported by the U.S. Department of Agriculture and used in the baseline analysis; total costs, profits and revenues are shown in Figure 3.

Other important parameters that affected the revenue stream were the REE concentration in spent FCC catalysts and REE bioleaching efficiency. Goonan reported that REE concentration in FCC catalysts ranged from 1.5% to 5.0% and averaged about 3.5%.[25] Leaching efficiencies determined from our work ranged from 28 to 56% depending upon pulp density and gluconic acid concentrations and could likely be further optimized. A range of these parameters for the bioleaching plant demonstrate that annual profits could vary significantly depending upon the FCC catalyst composition and leaching efficiency (Figure 4). The FCC catalyst used in this study was on the low end of the REE concentration range at 1.5% and the profit margin was low at this concentration. Assuming the 28% leaching efficiency from our study base case, the profit could increase from approximately $855,000 for 1.5% REE to $9,920,000 at 5% REE. Similarly, increasing the leaching efficiency would also lead to increases in profit. If a leaching efficiency of 50% could be achieved, the profits would potentially increase to approximately $4,160,000 at 1.5% REE to $20,900,000 at 5% REE.

The final set of parameters examined were the components of revenues for the bioleaching plant. First was the fraction of REE value attributed to the mixed REE leachate produced. As discussed in the Materials and Methods section, based upon the Mountain Pass Mine study,[26] 60% of the total costs to recover REE from ore is attributed to solvent extraction and separation steps while the remaining 40% is the cost to leach REE from the ore. On the basis of this, it was assumed that the value of mixed REE leachate was 40% of the value of purified REE. Given that the bioleachate described in this study could have an order of magnitude lower REE concentration than that obtained from a mining process (leading to higher costs for separation and concentration), this assumption may result in artificially high profits. A range of REE value percentages were examined from 0 to 40% at 2016 REE prices (Figure S3). The profit drops from $855,000/year to $738,000/year when the REE leachate has no value. This is only a 14% decrease which shows that this assumption has little effect on profits since most of the plant revenue is derived from the $200/ton tipping fee for the catalyst. If 2011 REE prices are considered, the

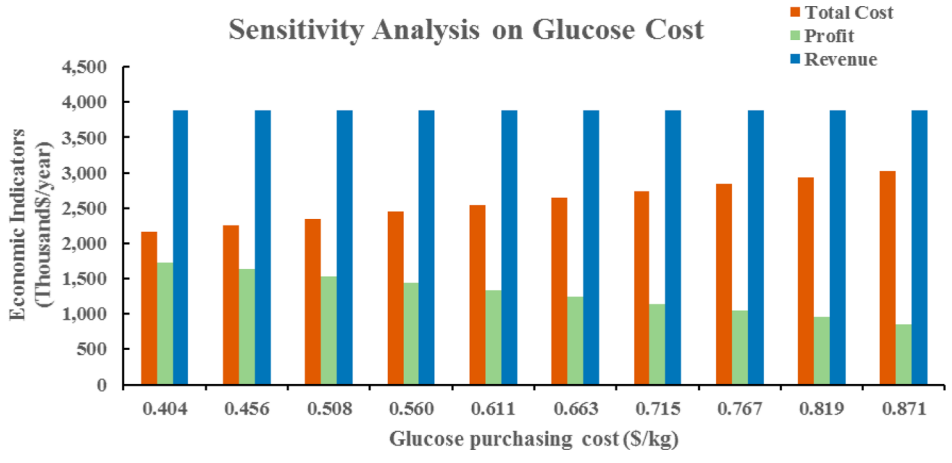


Figure 3. Sensitivity analysis on glucose purchase cost ($0.871/kg is the baseline glucose cost).

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

plant profit would be $6,260,000 if the mixed REE stream is valued at 40% of pure REE prices. In this case, if the mixed REE stream revenue is removed, the plant profits would drop by 88% since much more of the profit is attributed to REE revenue.

The next revenue parameter examined was the effect of the tipping fee on profits. The plant profits for a range of tipping fees (Figure S4) shows that a tipping fee of $155/ton is necessary for the plant to break even at 2016 REE prices. Contacts at Valero (Hude, personal communication) indicated that some of their spent catalyst is used as a cement additive, although no values for this application were available for consideration in the profit analysis. The final revenue parameter examined was the effect of REE price on plant profitability in the absence of a tipping fee (Figure S5). REE prices were varied from 5 to 27 times the 2016 prices and demonstrated that 26-fold increase over 2016 prices was the approximate breakeven point. Given that the primary REEs present in FCC catalyst were La and Ce, this was consistent with the plant profitability observed at 2011 prices where La and Ce were 50 times the 2016 prices.

Life Cycle Analysis. The results of the life cycle inventory (inputs and outputs shown in Table S7) analysis generated by TRACI to process 1 ton of spent FCC catalysts at a bioleaching facility show that the top three contributors to environmental impacts were electricity, glucose production, and steam generation (Table S8). As an example of the relative contributions of these factors, 66 kg equiv carbon dioxide (CO_2 equiv) would be produced to generate the required electricity while 28 kg CO_2 equiv would be emitted during production of glucose and 4.3 kg CO_2 equiv would be emitted to produce the required steam.

Figure 5 shows each direct process emission and input energy/material flow contribution to the bioleaching process,

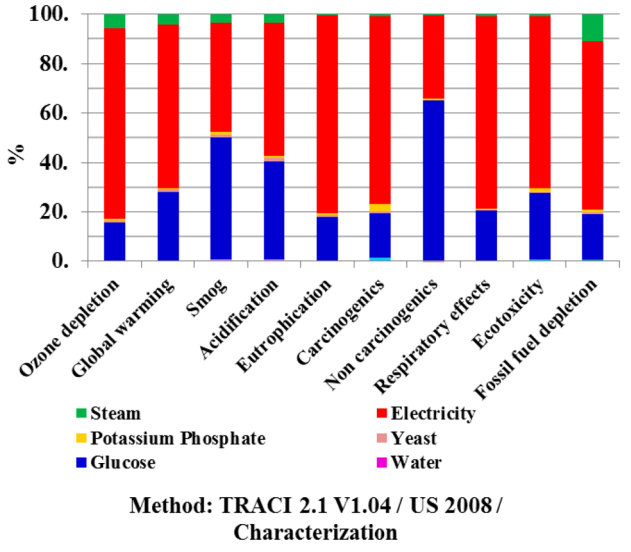


Figure 5. Contributions (% of total) of process emissions and energy/material flows at a bioleaching facility using TRACI.

where electricity and glucose combined dominated in all of the environmental impact categories. Electricity had the highest impact in all categories except smog and noncarcinogenics due to high electricity consumption for bioreactor agitation and the air compressor. Glucose had about the same impact as electricity in its contribution to smog and about twice the contribution to noncarcinogenics as electricity. Steam had a smaller impact with the largest being in fossil fuel depletion, but also had other impacts mainly due to the large amount of CO_2 released as the result of burning natural gas for steam generation and process heating. Potassium phosphate had minor but quantifiable impacts in several categories as well, with the largest being the carcinogenics category. Compared with glucose consumption in the bioleaching process, quantities of the other nutrients were insignificant and, therefore, showed very low impacts.

From Table 3, only the bioleaching process presented in this study and the nitric acid leaching process described by Gao and Owens[34] showed a positive profit. To determine if other factors could influence the choice of a chemical versus a biological leaching process, a comparative LCA was used to identify which process had the most benefits with regard to environmental impacts (Table 4). To ensure that the two

Table 4. Life Cycle Impacts for Bioleaching and Chemical Leaching Processes To Gain $4,000,000 Revenue from the Recycling Process Using TRACI

impact category	unit	bioleaching (A)	chemical leaching[34] (B)	(B)/(A)
ozone depletion	kg CFC-11 equiv	3.57×10^{-01}	1.04×10^{00}	2.9
global warming	kg CO_2 equiv	4.31×10^{06}	1.62×10^{07}	3.8
smog	kg O_3 equiv	1.45×10^{05}	6.48×10^{05}	4.5
acidification	kg SO_2 equiv	2.06×10^{04}	5.89×10^{04}	2.9
eutrophication	kg N equiv	4.69×10^{04}	4.57×10^{04}	1.0
carcinogenics	CTUh	2.71×10^{-01}	4.87×10^{-01}	1.8
noncarcinogenics	CTUh	3.21×10^{00}	2.19×10^{00}	0.7
respiratory effects	kg PM2.5 equiv	9.56×10^{03}	1.75×10^{04}	1.8
ecotoxicity	CTUe	3.05×10^{07}	6.47×10^{07}	2.1
fossil fuel depletion	MJ surplus	4.30×10^{06}	1.01×10^{07}	2.3

processes were compared on the same basis, the functional unit was defined as the resources required for each process to generate $4,000,000 in revenue. The nitric acid leaching process was found to have more deleterious impacts to the environment than bioleaching in all categories except eutrophication and noncarcinogenics. Smog and global warming categories were 4.5- and 3.8-fold higher (Table 4) for chemical leaching compared to bioleaching, and deleterious impacts from ozone depletion, acidification, fossil fuel depletion, ecotoxicity, respiratory effects, and carcinogenics ranging from 2.9- to 1.8-fold higher. The observed higher impacts for chemical leaching were due to increased electricity consumption while the glucose utilized in the bioleaching process led to the slightly higher impacts in the non-carcinogenics and eutrophication categories.

The TEA- and LCA-confirmed bioleaching has a high techno-economic potential compared to the alternative technologies while offering significant environmental benefits in most of the impact categories. These results suggest that further development of the bioleaching technique for REE extraction from spent FCC catalysts in a large-scale operation is warranted.

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.7b02771.

TEA assumptions for cost estimation, plant operation, financing, income tax, and depreciation; material costs; equipment and installation costs; other associated costs; LCA inventory data and impacts; process flow diagram and system boundaries; and sensitivity analyses (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: David.Reed@inl.gov.

ORCID

Vicki S. Thompson: 0000-0003-4975-392X
Yoshiko Fujita: 0000-0002-4472-4102
John W. Sutherland: 0000-0002-2118-0907
Yongqin Jiao: 0000-0002-6798-5823

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank J. Hude (Valero) for the FCC wastes and Masaaki Tazoe and Tatsuo Hoshino (NRL Pharma, Inc., Kawasaki, Japan) for their generous donation of 2,5-diketogluconic acid. We thank D. Daubaras (INL) and D. Lacroix (University of Idaho Center for Advanced Energy Studies) for assistance with HPLC and ICP analysis. We also thank Devin Imholte (INL) for helpful discussions and information provided for the TEA and David Combs (INL) for graphic. Hongyue Jin gratefully acknowledges support from the Environmental Research & Education Foundation Scholarship. We also acknowledge the DOE Science Undergraduate Laboratory Internship program for providing Van Nguyen's, Michael Jindra's, and Matthew Yim's funding. This research was supported by the Critical Materials Institute, an Energy Innovation Hub funded by the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Advanced Manufacturing Office, and conducted under DOE Idaho Operations Office Contract DE-AC07-05ID14517 and Lawrence Livermore National Laboratory Contract DE-AC52-07NA27344. Accordingly, the U.S. Government retains a nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, and allow others to do so, for U.S. Government purposes.

■ REFERENCES

(1) Binnemans, K.; Jones, P. T.; Blanpain, B.; Van Gerven, T.; Yang, Y. X.; Walton, A.; Buchert, M. Recycling of rare earths: a critical review. *J. Cleaner Prod.* **2013**, *51*, 1−22.

(2) Graedel, T. E.; Harper, E. M.; Nassar, N. T.; Reck, B. K. On the materials basis of modern society. *Proc. Natl. Acad. Sci. U. S. A.* **2015**, *112* (20), 6295−6300.

(3) DOE. *Critical Materials Strategy*; U.S. Department of Energy: Washington, D.C., Dec 2011.

(4) Nassar, N. T.; Du, X.; Graedel, T. E. Criticality of the Rare Earth Elements. *J. Ind. Ecol.* **2015**, *19* (6), 1044−1054.

(5) EPA. *U.S. Rare Earth Elements: A review of production, processing, recycling, and associated environmental issues*; U.S. Environmental Protection Agency: Washington, D.C., 2012.

(6) Haschke, M.; Ahmadian, J.; Zeidler, L.; Hubrig, T. In-Situ Recovery of Critical Technology Elements. *Procedia Eng.* **2016**, *138*, 248−257.

(7) Tunsu, C.; Petranikova, M.; Gergorić, M.; Ekberg, C.; Retegan, T. Reclaiming rare earth elements from end-of-life products: A review of the perspectives for urban mining using hydrometallurgical unit operations. *Hydrometallurgy* **2015**, *156*, 239−258.

(8) Diaz, L. A.; Lister, T. E.; Parkman, J. A.; Clark, G. G. Comprehensive process for the recovery of value and critical materials from electronic waste. *J. Cleaner Prod.* **2016**, *125*, 236−244.

(9) Priya, A.; Hait, S. Comparative assessment of metallurgical recovery of metals from electronic waste with special emphasis on bioleaching. *Environ. Sci. Pollut. Res.* **2017**, *24*, 6989−7008.

(10) Brierley, C. L.; Brierley, J. A. Progress in bioleaching: part B: applications of microbial processes by the minerals industries. *Appl. Microbiol. Biotechnol.* **2013**, *97* (17), 7543−7552.

(11) Zhuang, W.-Q.; Fitts, J. P.; Ajo-Franklin, C. M.; Maes, S.; Alvarez-Cohen, L.; Hennebel, T. Recovery of critical metals using biometallurgy. *Curr. Opin. Biotechnol.* **2015**, *33*, 327−335.

(12) Jain, N.; Sharma, D. K. Biohydrometallurgy for nonsulfidic minerals—a review. *Geomicrobiol. J.* **2004**, *21* (3), 135−144.

(13) Kermer, R.; Hedrich, S.; Bellenberg, S.; Brett, B.; Schrader, D.; Schönherr, P.; Köpcke, M.; Siewert, K.; Günther, N.; Gehrke, T.; Sand, W.; Räuchle, K.; Bertau, M.; Heide, G.; Weitkämper, L.; Wotruba, H.; Ludwig, H.-M.; Partusch, R.; Schippers, A.; Reichel, S.; Glombitza, F.; Janneck, E. Lignite ash: Waste material or potential resource - Investigation of metal recovery and utilization options. *Hydrometallurgy* **2017**, *168*, 141−152.

(14) Glombitza, F.; Reichel, S. Metal-Containing Residues from Industry and in the Environment: Geobiotechnological Urban Mining. In *Geobiotechnology I*; Schippers, A., Glombitza, F., Sand, W., Eds.; Springer: Berlin, 2014; Vol. *141*, pp 49−107.

(15) Shin, D.; Kim, J.; Kim, B.-s.; Jeong, J.; Lee, J.-c. Use of Phosphate Solubilizing Bacteria to Leach Rare Earth Elements from Monazite-Bearing Ore. *Minerals* **2015**, *5* (2), 189−202.

(16) Brisson, V. L.; Zhuang, W. Q.; Alvarez Cohen, L. Bioleaching of rare earth elements from monazite sand. *Biotechnol. Bioeng.* **2016**, *113* (2), 339−348.

(17) Hassanien, W. A.; Desouky, O. A.; Hussien, S. S. Bioleaching of some rare earth elements from Egyptian monazite using *Aspergillus ficuum* and *Pseudomonas aeruginosa*. *Walailak J.Sci. Technol. (WJST)* **2014**, *11* (9), 809−823.

(18) Reed, D. W.; Fujita, Y.; Daubaras, D. L.; Jiao, Y.; Thompson, V. S. Bioleaching of rare earth elements from waste phosphors and cracking catalysts. *Hydrometallurgy* **2016**, *166*, 34−40.

(19) Amiri, F.; Yaghmaei, S.; Mousavi, S. M. Bioleaching of tungsten-rich spent hydrocracking catalyst using *Penicillium simplicissimum*. *Bioresour. Technol.* **2011**, *102* (2), 1567−1573.

(20) Qu, Y.; Lian, B. Bioleaching of rare earth and radioactive elements from red mud using *Penicillium tricolor* RM-10. *Bioresour. Technol.* **2013**, *136* (0), 16−23.

(21) Maes, S.; Zhuang, W.-Q.; Rabaey, K.; Alvarez-Cohen, L.; Hennebel, T. Concomitant Leaching and Electrochemical Extraction of Rare Earth Elements from Monazite. *Environ. Sci. Technol.* **2017**, *51* (3), 1654−1661.

(22) Corbett, M. K.; Eksteen, J. J.; Niu, X.-Z.; Croue, J.-P.; Watkin, E. L. J. Interactions of phosphate solubilising microorganisms with natural rare-earth phosphate minerals: a study utilizing Western Australian monazite. *Bioprocess Biosyst. Eng.* **2017**, *40* (6), 929−942.

(23) Asai, T.; Shoda, K. The taxonomy of *Acetobacter* and allied oxidative bacteria. *J. Gen. Appl. Microbiol.* **1958**, *4* (4), 289−311.

(24) Pikovskaya, R. Mobilization of phosphorus in soil in connection with vital activity of some microbial species. *Mikrobiologiya* **1948**, *17*, 362−370.

(25) Goonan, T. G. *Rare earth elements: End use and recyclability*; 2328-0328; U.S. Geological Survey: Washington, D.C., 2011.

(26) Consulting, S. *Alternative Technical Economic Model for the Mountain Pass Re-Start Project*; https://www.sec.gov/Archives/edgar/

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609

data/1489137/000095012310065239/d74323fwfwp.htm#164, 2010 (accessed March 8, 2017).
(27) Efe, Ç.; van der Wielen, L. A.; Straathof, A. J. Techno-economic analysis of succinic acid production using adsorption from fermentation medium. *Biomass Bioenergy* **2013**, *56*, 479−492.
(28) Silla, H. *Chemical process engineering: design and economics*; CRC Press: Boca Raton, FL, 2003.
(29) Peters, M. S. *Plant design and economics for chemical engineers*; Prentice Hall: Upper Saddle River, NJ, 1991; Vol. *4*.
(30) Marafi, M.; Stanislaus, A. Spent hydroprocessing catalyst management: A review: Part II. Advances in metal recovery and safe disposal methods. *Resour., Conserv.Recy.*. **2008**, *53* (1), 1−26.
(31) Aung, K. M. M.; Ting, Y.-P. Bioleaching of spent fluid catalytic cracking catalyst using Aspergillus niger. *J. Biotechnol.* **2005**, *116* (2), 159−170.
(32) Bare, J. TRACI 2.0: the tool for the reduction and assessment of chemical and other environmental impacts 2.0. *Clean Technol. Environ. Policy* **2011**, *13* (5), 687−696.
(33) Silberbach, M.; Maier, B.; Zimmermann, M.; Büchs, J. Glucose oxidation by *Gluconobacter oxydans*: characterization in shaking-flasks, scale-up and optimization of the pH profile. *Appl. Microbiol. Biotechnol.* **2003**, *62* (1), 92−98.
(34) Gao, X.; Owens, W. T. Process for Metal Recovery from Catalyst Waste. WO2012/082597 A2, 2012.
(35) Innocenzi, V.; Ferella, F.; De Michelis, I.; Vegliò, F. Treatment of fluid catalytic cracking spent catalysts to recover lanthanum and cerium: Comparison between selective precipitation and solvent extraction. *J. Ind. Eng. Chem.* **2015**, *24*, 92−97.
(36) Zhao, Z.; Qiu, Z.; Yang, J.; Lu, S.; Cao, L.; Zhang, W.; Xu, Y. Recovery of rare earth elements from spent fluid catalytic cracking catalysts using leaching and solvent extraction techniques. *Hydrometallurgy* **2017**, *167*, 183−188.

DOI: 10.1021/acssuschemeng.7b02771
ACS Sustainable Chem. Eng. 2018, 6, 1602−1609